

Multi-Branch Retail Performance & Operations Intelligence

Project Summary Document

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Tools: Excel • MySQL • Power BI • Google Docs • Google Slides • GitHub

Item	Details
Project	Multi-Branch Retail Performance & Operations Intelligence
Geography	Mumbai, Pune, Delhi, Bangalore
Tools	Excel, MySQL 8.0, SQL, Power BI, DAX
Dashboards	5
Validation	Dashboard KPI/chart outputs reconciled against SQL outputs during development
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Business Problem

The retail business operates across **Mumbai, Pune, Delhi, and Bangalore** and generates large volumes of sales, customer, product, shipment, payment, and return data. Management needs a centralized view of performance to identify growth opportunities, customer and product priorities, and operational issues.

Business Task

Analyze retail performance across **sales, stores, products, customers, shipments, returns, and refunds**. Compare the four cities, identify key performance drivers, measure customer and operational metrics, validate results using SQL, and provide actionable recommendations.

Project Objective

- Analyze sales, growth and AOV.
- Compare performance across cities.
- Identify top products and categories.
- Measure customer value and repeat behavior.
- Evaluate shipment and return performance.

- Build interactive Power BI dashboards.
- Validate key metrics against SQL.
- Generate actionable business insights and recommendations.

Dataset

The analysis uses a 12-table retail data warehouse dataset containing approximately 1.59 million records across the warehouse tables.

Table	Number of Records
Categories	30
Customers	50,000
Employee	1,000
Order Items	600,000
Orders	300,000
Payments	300,000
Products	10,000
Promotions	50
Returns	30,000
Shipments	300,000
Stores	100
Suppliers	200
Total	1,591,380

Technology Stack

Tool	Purpose
Excel	Initial data exploration
MySQL 8.0	Data warehouse querying, EDA, aggregation and validation
SQL	EDA, joins, aggregations, distinct counts, KPI calculations and validation
Power BI	Data modeling, relationships, slicers and dashboard
DAX	Reusable measures, ratios and calculated columns
Google Docs	Project documentation
Google Slides	Project presentation
GitHub	Portfolio hosting and version control

Analytics Workflow

Raw Retail Data → Initial Data Inspection → MySQL/SQL Analysis → SQL KPI Validation → Power BI Data Model → DAX Measures → Dashboards → SQL Reconciliation → Business Insights → Recommendations.

Data Model

Core analytical flow:

- Customers → Orders → Order Items → Returns
- Orders → Shipments

Relationship and filter propagation were tested during development, particularly for city-level shipment and return analysis.

Consolidated Validated KPI Register

Dashboard/Area	Metric	Validated Value
D1 Executive	Gross Sales	~\$3.83B
D1 Executive	YoY Sales Growth	29.77%
D1 Executive	AOV	₹12.76K
D2 Store	Top Market	Mumbai
D2 Store	Mumbai Sales Contribution	~31%
D3 Product	Top Product	745
D3 Product	Top Category	Cat_5
D3 Product	Distinct Products	~10K
D3 Product	Total Quantity	~2M
D4 Customer	Total Customers	~50K
D4 Customer	Active Customers	49,886
D4 Customer	Repeat Customers	49,168
D4 Customer	One-Time Customers	718
D4 Customer	Repeat Rate	98.56%
D4 Customer	Avg Orders/Customer	6.01
D4 Customer	Avg Customer Value	₹76.73K
D4 Customer	Top Customer	33455
D5 Operations	Total Shipments	300,000
D5 Operations	Late Shipments	99,855
D5 Operations	Late Delivery Rate	33.29%
D5 Operations	Refunds	₹75,962,300
D5 Operations	Returned Items	29,251

Final Business Story

The four-city retail business generated approximately ₹3.83B in gross sales with 29.77% year-over-year growth. Mumbai is the strongest market across sales, orders and customer value, while customer retention is exceptionally strong at 98.56%. The largest growth opportunity is therefore not simply customer acquisition, but increasing the value of existing customers.

Product performance is diversified across approximately 10K products, with Cat_5 and Product 745 leading their respective rankings. Operationally, the most significant concern is delivery reliability, with 33.29% of shipments classified as late, alongside ₹75.96M in refunds and 29,251 returned items.

Management should prioritize delivery improvement, reduce avoidable returns, protect high-performing products and markets, and use customer-value strategies to drive further revenue growth.

Conclusion

The Multi-Branch Retail Performance & Operations Intelligence Project successfully transforms retail data into a structured, validated business intelligence solution covering sales, store performance, products and categories, customers, shipments, returns, and refunds across Mumbai, Pune, Delhi, and Bangalore.

Using Excel for initial data inspection, MySQL/SQL for detailed analysis and independent validation, and Power BI with DAX for data modeling and dashboard development, the project delivers five interactive dashboards that provide a comprehensive view of business and operational performance.

All major Power BI KPIs and analytical outputs were validated against SQL results, improving confidence in the reported metrics, while documented data limitations ensure conclusions remain within what the dataset can reliably support.

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